

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 14

DATE : 26-12-2019

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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REVISION TEST # 14

Topics

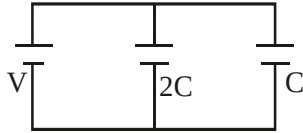
DATE : 26 - 12 - 2019**PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]****PHYSICS : Current Electricity and Capacitors, Moving Charges and Magnetism****CHEMISTRY : Hydrocarbon, Aromatic Compound****BOTANY : Plant Anatomy(Meristem,Perenchyma,Vascular Tissue),Biotech-II**

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

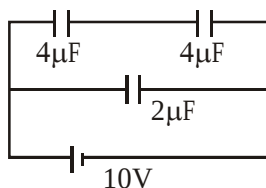
BEWARE OF NEGATIVE MARKING

1. Two condensers, one of capacity C and the other capacity $2C$, are connected to a V volt battery, as shown in figure



The work done in charging fully both the condenser is

- (1) $\frac{1}{2}CV^2$ (2) $2CV^2$
(3) $\frac{3}{2}CV^2$ (4) $3CV^2$
2. If initial charge on all the capacitors were zero, work done by the battery in the circuit shown is-



- (1) 0.2 mJ (2) 200 mJ
(3) 0.4 mJ (4) 400 mJ
3. The energy required to charge a parallel plate condenser of plate separation d and plate area of cross-section A , such that the surface charge density on the plate will be σ , is

- (1) $\frac{A\sigma^2 d}{2\epsilon}$ (2) $\frac{A\sigma^2 d}{\epsilon}$
(3) $\frac{\epsilon\sigma^2 d}{2}$ (4) $\frac{A\epsilon\sigma^2}{d}$

4. Three identical capacitors are connected in series. The capacitance of combination is C and break-down voltage is V . The net capacitance and break down voltage when these are connected in parallel combination will be :

(1) $\frac{C}{3}, 3V$

(2) $9C, \frac{V}{3}$

(3) $\frac{C}{3}, V$

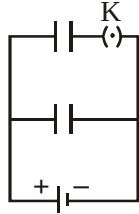
(4) $3C, \frac{V}{3}$

5. A parallel plate capacitor of capacitance C is connected to a cell of emf V . A dielectric slab of dielectric constant K , that can just fill the air gap of capacitor, is now inserted in it. Which of the following is incorrect?

- (1) The energy stored in the capacitor increases K times
(2) The change in energy stored is $\frac{1}{2}CV^2(K-1)$
(3) The charge on the capacitor remain same as before
(4) The potential difference between the plates remain unchanged.

SPACE FOR ROUGH WORK

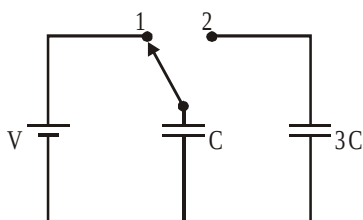
6. Two identical parallel plate capacitors are connected as shown in the circuit. The key K is unplugged and identical dielectric material is filled in both capacitors. Choose the correct statement.



- (1) Both capacitor will have same charge
(2) Both capacitor will have same potential
(3) Both capacitor will have same energy stored.
(4) The ratio of energy stored in capacitors will be $1:K^2$.
7. A parallel plate capacitor has uniform electric field E between the plates and surface area of each plate is A . The force of attraction between the plates of the capacitor is

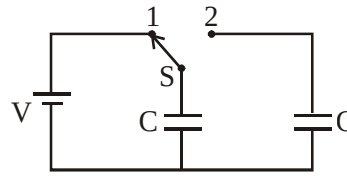
- (1) $E^2 A \epsilon_0$ (2) $2E^2 A \epsilon_0$
(3) $\frac{E^2 A \epsilon_0}{2}$ (4) $\frac{E^2}{2A \epsilon_0}$

8. Two capacitors of capacitance C and $3C$ are shown in the diagram. When switch S turned to position 2, the percentage of stored energy dissipated is



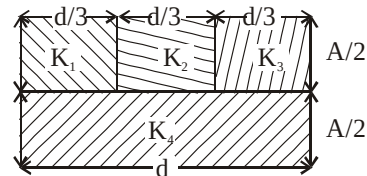
- (1) 0% (2) 25%
(3) 75% (4) 100%

9.



Two identical capacitors as shown in the diagram. when switch S is turned to position 2, then choose incorrect statement :-

- (1) Charge on both capacitor will be same
(2) Potential difference across each capacitor will be half
(3) Energy stored in capacitors remain unchanged.
(4) Energy stored in capacitors will be reduced to half.
10. A parallel plate capacitor of area A , plate separation d and capacitance C is filled with four dielectric material having dielectric constant K_1, K_2, K_3 and K_4 as shown in the figure. If a single dielectric material is to be used to have the same capacitance in this capacitor, then its dielectric constant K is given by:-



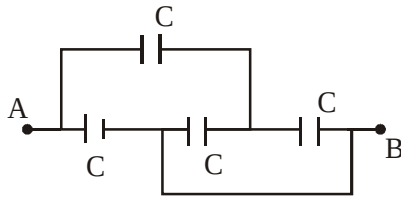
- (1) $\frac{2}{K} \pm \frac{3}{K_1 + K_2 + K_3} + \frac{1}{K_4}$
(2) $\frac{1}{K} = \frac{1}{K_1} + \frac{1}{K_2} + \frac{1}{K_3} + \frac{3}{2K_4}$
(3) $K = \frac{3K_1 K_2 K_3}{2(K_2 K_3 + K_1 + K_2 + K_3 K_1)} + \frac{K_4}{2}$
(4) $K = \frac{2}{3}(K_1 + K_2 + K_3) + 2K_4$

SPACE FOR ROUGH WORK

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| <p>11. A capacitor of capacitance $2\mu\text{F}$ is charged by a battery. The battery is removed and another uncharged capacitor of capacitance $6\mu\text{F}$ is connected in parallel. The total electrostatic energy of resulting system :-</p> <ol style="list-style-type: none"> (1) Decrease by a factor of 4 (2) Remains the same (3) Increases by a factor of 4 (4) Increases by a factor of 2 <p>12. Select incorrect statement for the electrostatic force between the metal plates of an isolated charged parallel plate capacitor :</p> <ol style="list-style-type: none"> (1) Inversely proportional to the area of plates (2) Directly proportional to square of charge on the plates (3) Independent of the distance between the plates (4) Directly proportional to potential difference between two plates <p>13. If separation between two plates of isolated charged parallel plate capacitor is doubled, then</p> <ol style="list-style-type: none"> (1) The electrostatic force between two plates will be 4 times (2) The potential difference between two plates will be doubled. (3) Charge on the plates will be doubled. (4) Electric field between two plates will be doubled <p>14. Three capacitors each of capacitance C were to be connected in such a way that the effective capacitance becomes $\frac{3C}{2}$.</p> | <p>This can be done by:-</p> <ol style="list-style-type: none"> (1) Connecting all of them in series (2) Connecting then in parallel (3) Connecting two in series and one in parallel (4) Connecting two in parallel and one in series <p>15. Twenty four capacitors each of capacitance $2\mu\text{F}$ are to be connected in such a way that the effective capacitance of $3\mu\text{F}$. This can be done by :-</p> <ol style="list-style-type: none"> (1) Connecting 6 in series (in each row) and 4 rows in parallel. (2) Connecting 8 in series (in each row) and 3 rows in parallel (3) Connecting 3 in series (in each row) and 8 rows in parallel (4) Connecting 4 in series (in each row) and 6 rows in parallel <p>16. A parallel plate air capacitor is charged by a battery. After disconnecting the battery, the distance between the plates of the capacitor is increased using an insulating handle. Choose the incorrect statement about this</p> <ol style="list-style-type: none"> (1) Electric field between plates remain unchanged (2) Potential difference between plates increases (3) Electrostatic force between plates decreases. (4) Electrostatic energy density remain unchanged. |
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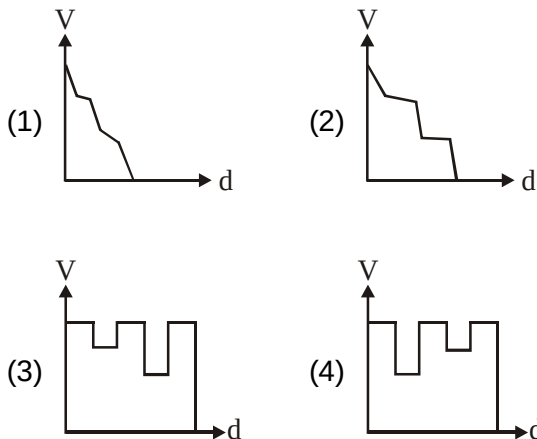
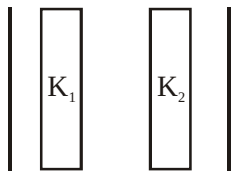
SPACE FOR ROUGH WORK

17. The effective capacity of the network between terminals A and B in

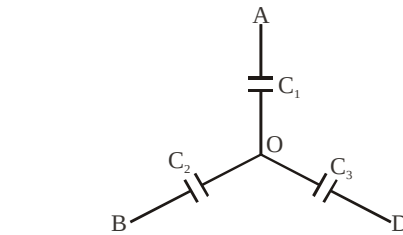


- (1) C (2) $\frac{3C}{2}$ (3) $\frac{5C}{3}$ (4) $\frac{3C}{5}$

18. A parallel plate capacitor is given. Choose most appropriate graph of potential of it. ($K_1 > K_2$) (Where K represent dielectric constant)

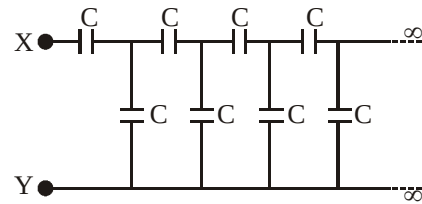


19. Three uncharged capacitors of capacitance C_1 , C_2 and C_3 are connected as shown in figure to one another and to points A, B and D at potentials V_A , V_B and V_D . Then the potential at O will be :

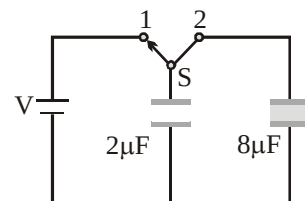


- (1) $\frac{V_A + V_B + V_D}{C_1 + C_2 + C_3}$
(2) $\frac{V_A C_1 + V_B C_2 + V_D C_3}{C_1 + C_2 + C_3}$
(3) $\frac{V_A V_B + V_B V_D + V_D V_A}{C_1 + C_2 + C_3}$
(4) $\frac{V_A V_B V_D}{C_1 C_2 + C_2 C_3 + C_3 C_1}$

20. In the adjoining figure, the effective capacity of the group of condensers will be :-



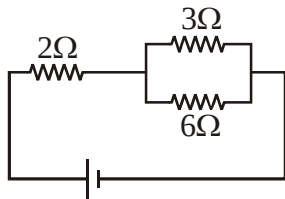
- (1) $2C$ (2) ∞
(3) zero (4) $0.62 C$
21. A $2\mu F$ capacitor is charged as shown in figure. The percentage of its stored energy dissipated after the switch S is turned to position 2 is:-



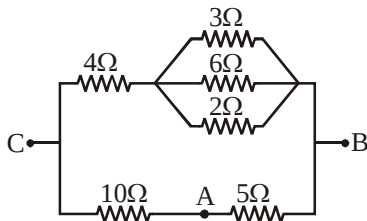
- (1) 0% (2) 20%
(3) 75% (4) 80%

SPACE FOR ROUGH WORK

22. A current of 2A flows through 6Ω resistance shown in figure. The power dissipated in the 2Ω resistor is :-

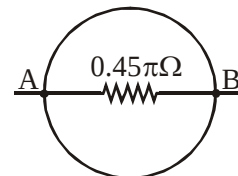


- (1) 36 W (2) 18 W
(3) 72 W (4) 144 W
23. A Galvanometer of resistance 100Ω is connected to a battery of 5V along with a resistance of 4900Ω in series. A full scale deflection of 40 division is obtained in galvanometer. In order to reduce this deflection to 20 division, the additional resistance in series should be :-
- (1) 1000Ω (2) 9900Ω
(3) 5000Ω (4) 10000Ω
24. The product of resistivity and conductivity of material is :-
- (1) zero (2) Infinity
(3) unity (4) None
25. In the circuit shown, the current through the 2Ω resistor is 1A. The potential difference between points A and B is :-



- (1) $\frac{5}{3}V$ (2) $\frac{10}{3}V$ (3) $\frac{20}{3}V$ (4) $\frac{25}{3}V$

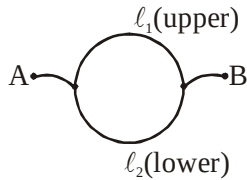
26. An electric kettle takes 2A current at 200 V. Then how much time will it take to boil 2kg of water from temperature 40°C . The temperature of boiling water is 100°C :-
- (1) 7 min
(2) 14 min
(3) 21 min
(4) 35 min
27. A heater coil takes 4 min to boil some amount of water. Another heater will takes 6 min to boil the same amount of water with the same supply voltage. Now the two coils are connected in parallel combination then how much time will it take to boil the same amount of water. (Supply voltage is same in all cases) :-
- (1) 1.2 min (2) 2.4 min
(3) 3.6 min (4) 4.8 min
28. A wire of resistance 18 ohm per meter is bent to form a complete circle of radius 10 cm. A wire of 0.45π ohm is connected between two diametrically opposite points A and B as shown in figure then net resistance between A and B is :-



- (1) $0.6\pi\Omega$
(2) $0.9\pi\Omega$
(3) $0.3\pi\Omega$
(4) $1.2\pi\Omega$

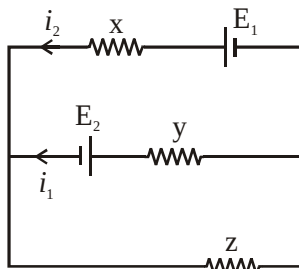
SPACE FOR ROUGH WORK

29. A wire of resistance 12 ohm is bent into a circular loop. Two points A and B are taken such that the length ratio of upper and lower part is 1 : 3. The net resistance between points A and B is :-



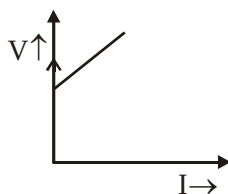
- (1) 2.25 Ω (2) 4.5 Ω
(3) 6.75 Ω (4) 9 Ω

30. For the circuit shown in figure, which one of the following equation is correct :-



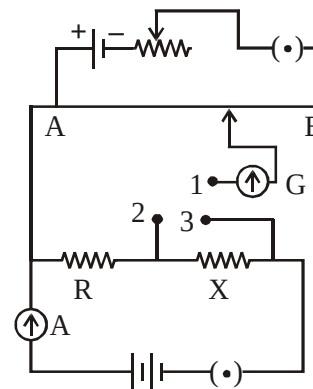
- (1) $E_1 - i_2 X + E_2 + i_1 Y = 0$
(2) $E_1 - i_2 X - i_1 Z = 0$
(3) $E_2 + i_2 Y - i_2 Z = 0$
(4) $-E_2 + (i_1 + i_2)Z + i_1 Y = 0$

31. For the given figure, graph is drawn in between terminal potential difference (V) of a cell of emf E and of internal resistance r as function of current (I), flowing through it, during charging of the cell. The slope of graph and intercept are respectively :-



- (1) $-E$ and r
(2) E and r
(3) $-r$ and E
(4) r and E

32. A potentiometer circuit set up is shown in figure. The potential gradient across the potentiometer wire is K V/cm and the ammeter, present in the circuit reads 1A when two way key is switched off. The balance points, when the key between the terminals (i) 1 and 2 (ii) 1 and 3 is plugged in, are found to be at lengths ℓ_1 cm and ℓ_2 cm respectively. Then the ratio of magnitude of the resistors R and X in ohms is equal to :-



- (1) ℓ_1 / ℓ_2
(2) $(\ell_2 - \ell_1) / \ell_2$
(3) $\ell_1 / (\ell_2 - \ell_1)$
(4) $(\ell_2 - \ell_1) / \ell_1$

SPACE FOR ROUGH WORK

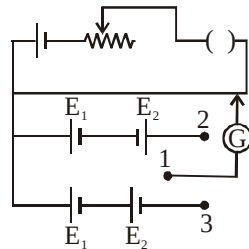
33. A potentiometer setup is shown in figure. The balance point when the key between the terminals (i) 1 and 2 (ii) 1 and 3 is plugged in, are found to be at lengths x cm and y cm respectively. If potential gradient is k V/cm then the emf E_1 and E_2 are :-

(1) $\frac{k}{2}(x+y), \frac{k}{2}(y-x)$

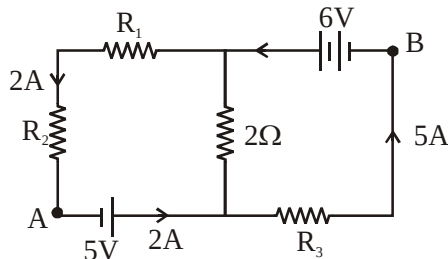
(2) $\frac{k}{2}(2x+y); \frac{k}{2}(2y-x)$

(3) $\frac{k}{2}(x-y); \frac{k}{2}(x+y)$

(4) $\frac{k}{2}(y-x); \frac{k}{2}(x+y)$



34. In the circuit shown in the figure, if the potential at point A is taken to be $-2V$, the potential at point B is :-



- (1) $-3V$ (2) $3V$ (3) $-4V$ (4) $5V$

35. A galvanometer of resistance G is shunted by a resistance S ohm. To keep the main current half to initial value in the galvanometer, the resistance to be put in series with the galvanometer is :-

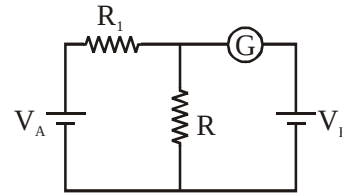
(1) $\frac{G(G+2S)}{(G+S)}$

(2) $\frac{G(2G+S)}{(G+S)}$

(3) $\frac{S(G+2S)}{(G+S)}$

(4) $\frac{S(2G+S)}{(G+S)}$

36. In the circuit shown the cells A and B have negligible resistances. for $V_A = 16V$, $R_1 = 600\Omega$ and $R = 200\Omega$, the galvanometer shows no deflection. The value V_B is :-



(1) $12V$

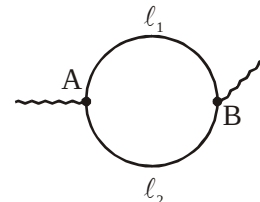
(2) $4V$

(3) $16V$

(4) $8V$

37. A ring is made of a wire having a resistance $R_0 = 15\Omega$. Find the points A and B as shown in figure at which a current carrying conductor should be connected so that the resistance R of the sub circuit between these point is equal

to $\frac{10}{3}\Omega$:-



(1) $\frac{l_1}{l_2} = \frac{1}{2}$

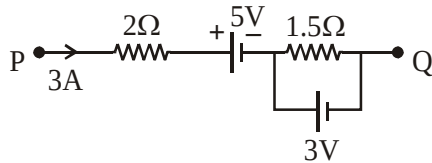
(2) $\frac{l_1}{l_2} = \frac{6}{5}$

(3) $\frac{l_1}{l_2} = \frac{4}{5}$

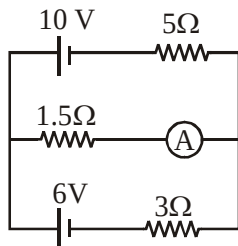
(4) $\frac{l_1}{l_2} = \frac{1}{4}$

SPACE FOR ROUGH WORK

38. The potential difference ($V_P - V_Q$) between points P and Q in the given figure is :-

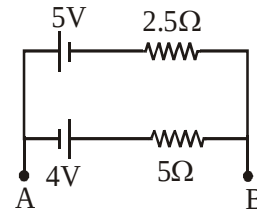


- (1) 7 V (2) 21 V
(3) 14 V (4) 28 V
39. The resistance of a wire is 32Ω . Now it is stretched till its thickness is reduced by 20%. The new resistance of the wire is :-
- (1) 25 Ω (2) 50 Ω
(3) 32 Ω (4) 64 Ω
40. A potentiometer is preferred over voltmeter while measuring voltage because :-
- (1) It doesn't draw any current from main circuit
(2) its resistance is high
(3) its resistance is low
(4) can't be predicted.
41. In the given circuit, reading of ammeter is nearly:-



- (1) 8 A (2) 5A
(3) 1A (4) 2A

42. In the given figure if the circuit is reduced into a single cell then emf and internal resistance of equivalent cell is :-



- (1) $\frac{14}{3}$ V, 7.5Ω (2) $\frac{14}{3}$ V, $\frac{5}{3}\Omega$
(3) 2V, $\frac{5}{3}\Omega$ (4) 2V, 7.5Ω
43. A cell of 5V emf and 0.75Ω internal resistance delivers a current of 3A in an external load then the terminal potential difference of the cell is :-
- (1) 2.75 V (2) 7.25 V
(3) 3.5 V (4) 5.75 V
44. Resistance per unit length of potentiometer wire is $10^{-4} \Omega/\text{m}$ and current flowing through it is 10A, then the potential gradient will be :-
- (1) 0.1 mV/m (2) 10 mV/m
(3) 1 mV/m (4) 100 mV/m
45. 20 identical cell each of emf E and internal resistance 'r' are connected with a load resistance R in series combination. Current flow in the circuit is I. Now n cells are connected with opposite polarities then the current flow in the circuit is I/2. Then the value of n is :-
- (1) 5 (2) 10 (3) 15 (4) 20

SPACE FOR ROUGH WORK

46. Which of the following alkyl halides gives a mixture of alkenes on dehydrohalogenation

- (1) n-Propyl halide
- (2) Isopropyl halide
- (3) s-Butyl bromide
- (4) t-Butyl bromide

47. Hydroboration of isobutylene followed by oxidative alkaline hydrolysis gives

- (1) 1-butanol
- (2) 2-butanol
- (3) Isobutyl alcohol
- (4) $(\text{CH}_3)_3\text{C}-\text{OH}$

48. Ozonolysis of 3-Methyl-1-butene gives a mixture of

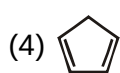
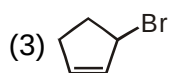
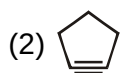
- (1) Propanal and ethanal
- (2) Propanone and ethanal
- (3) 2-Methylpropanal and methanal
- (4) Butanone and methanal

49. Oxidation of isobutylene with acid potassium permanganate gives

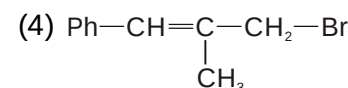
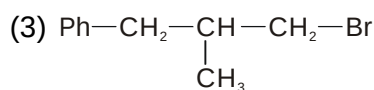
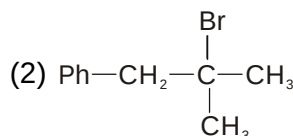
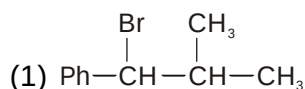
- (1) Acetone + CO_2
- (2) Acetic acid
- (3) Acetic acid + CO_2

(4) Acetic acid + acetone

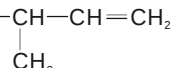
50.  $\xrightarrow{\text{Br}_2}$ A $\xrightarrow{\text{alc KOH}}$ B ; What is the structure of B



51. $\text{Ph}-\text{CH}=\text{C}(\text{CH}_3)-\text{CH}_3 + \text{HBr} \xrightarrow{\text{Peroxide}} (\text{X})$
'X' is



52. Which of the following is maximum reactive for NBS

- (1) PhCH_3
- (2) $\text{PhCH}_2-\text{CH}_3$
- (3) $\text{PhCH}_2-\text{CH}=\text{CH}_2$
- (4) 

53. The correct order of increasing stability of the given alkenes is

- (1) 1-pentene > cis-2-pentene > trans-2-pentene > 2-methyl-2-butene
- (2) 1-pentene > trans-2-pentene > cis-2-pentene > 2-methyl-2-butene
- (3) 1-pentene < cis-2-pentene < trans-2-pentene < 2-methyl-2-butene
- (4) 1-pentene < trans-2-pentene < cis-2-pentene < 2-methyl-2-butene

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54. The order of reactivity of alkyl halides in Wurtz reaction is
 (1) $R-I > R-Br > R-Cl$
 (2) $R-I < R-Br < R-Cl$
 (3) $R-Br > R-I < R-Cl$
 (4) $R-I > R-Cl > R-Br$
55. Wurtz reaction on a mixture of ethyl halide and isobutyl halide gives
 (1) Butane and isobutane
 (2) Butane and 2,5-dimethylhexane
 (3) Butane, 2,5-dimethylhexane and isohexane
 (4) Butane and isohexane
56. The Corey-House alkane synthesis is carried out by treating an alkyl halide with
 (1) Lithium metal
 (2) Copper metal
 (3) Lithium metal followed by reaction with cuprous iodide and then treating the product with an alkyl halide
 (4) Cuprous iodide followed by reaction with alkyl halide
57. Which of the following acids on decarboxylation gives isobutane
 (1) 2,2-Dimethyl butanoic acid
 (2) 2,2-dimethyl propanoic acid
 (3) 3-Methyl pentanoic acid
 (4) 2-Methyl butanoic acid
58. The volume of methane evolved by treatment of 16.6 g of methyl magnesium iodide with water at S.T.P. is [Atomic Weight of I=127]
 (1) 224 mL (2) 2.24 L
 (3) 0.224 L (4) 22.4 L
59. Give reactivity order for decarboxylation ?
 CH_3-CH_2-COOH ; (I)
 $CH_2=CH-COOH$; (II)
 $CH \equiv C-COOH$ (III)
 (1) $I > II > III$ (2) $III > II > I$
 (3) $III > I > II$ (4) $II > I > III$
60. $(CH_3)_3C-Br \xrightarrow[(ii) CuI]{(i) Li} A \xrightarrow{CH_3CH_2CH_2Br} B$;
 'B' is
 (1) $CH_3(CH_2)_2C(CH_3)_3$
 (2) $(CH_3)_3C-C(CH_3)_3$
 (3) $CH_3(CH_2)_4CH_3$
 (4) $(CH_3)_2CH_2C(CH_3)_3$
61. Which of the following reactions does not involve a C—C bond formation
 (1) Hydrolysis of a Grignard reagent
 (2) Combination of two alkyl free radicals
 (3) Corey-House synthesis of alkanes
 (4) $RNa + R-Br \longrightarrow R-R + NaBr$
62. The highest boiling point is expected for
 (1) Isooctane
 (2) 2,2,3,3-tetramethylbutane
 (3) n-octane
 (4) n-butane
63. Which alkane is liquid at room temperature
 (1) Hexane (2) Butane
 (3) Propane (4) Ethane
64. Alkane reacts with which of the following halogens in dark
 (1) F_2 (2) Cl_2
 (3) I_2 (4) Br_2

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- | | |
|---|---|
| <p>65. Pyrolysis of alkanes is a</p> <ol style="list-style-type: none"> (1) Nucleophilic addition reaction (2) Free radical substitution reaction (3) Electrophilic addition reaction (4) Free radical elimination reaction <p>66. The nitrating agent for the nitration of alkanes is:</p> <ol style="list-style-type: none"> (1) Conc. HNO_3 (2) Mixture of conc. HNO_3 and conc. H_2SO_4 (3) Acetyl nitrate (4) HNO_3 vapours at high temperature <p>67. The antiknock compound is</p> <ol style="list-style-type: none"> (1) TEL (2) Diethylzinc (3) Dimethylcadmium (4) Tetramethyl tin <p>68. The chain propagating step is fastest in the reaction of an alkane with</p> <ol style="list-style-type: none"> (1) Fluorine free radical (2) Chlorine free radical (3) Iodine free radical (4) Bromine free radical <p>69. 10 litre of an alkane X require 35 litre of O_2 for complete combustion. X forms only one monochloro derivative Y. The action of alcoholic KOH on Y yields</p> <ol style="list-style-type: none"> (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (2) $\text{CH}_2=\text{CH}_2$ (3) $\text{CH}_3\text{CH}=\text{CH}_2$ (4) $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$ | <p>70. The compound with the highest boiling point is</p> <ol style="list-style-type: none"> (1) n-hexane (2) n-pentane (3) 2,2-dimethyl propane (4) propane <p>71. Photochemical chlorination of alkane is initiated by a process of</p> <ol style="list-style-type: none"> (1) pyrolysis (2) substitution (3) Homolysis (4) Peroxidation <p>72. Isomerization in alkane may be brought about by using :</p> <ol style="list-style-type: none"> (1) Al_2O_3 (2) Fe_2O_3 (3) AlCl_3 and HCl (4) Concentrated H_2SO_4 <p>73. Bromination of an alkane as compared to chlorination proceeds</p> <ol style="list-style-type: none"> (1) At a slower rate (2) At a faster rate (3) With equal rates (4) With equal or different rate depends upon the temperature <p>74. The pH of solution in Kolbe's electrolysis</p> <ol style="list-style-type: none"> (1) increase with time (2) decrease with time (3) remains constant (4) nothing can be said <p>75. Which of the following compound will not give a precipitate with Tollen's reagent</p> <ol style="list-style-type: none"> (1) ethyne (2) 1-butyne (3) 3-methyl -1-butyne (4) 1-pentene |
|---|---|

SPACE FOR ROUGH WORK

76. The reaction $\text{CH}\equiv\text{CH} \longrightarrow \text{CHCl}_2\text{CHO}$ takes place in

- (1) $\text{CH}\equiv\text{CH} \xrightarrow{\text{HOCl}}$
- (2) $\text{HC}\equiv\text{CH} \xrightarrow{\text{HCl / peroxide}}$
- (3) $\text{CH}\equiv\text{CH} \xrightarrow{\text{Cl}_2}$
- (4) $\text{CH}\equiv\text{CH} \xrightarrow{\text{Cl}_2 / \text{oxidizing agent}}$

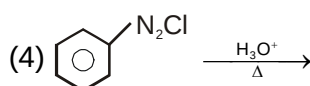
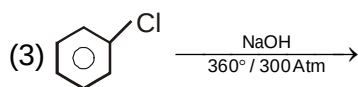
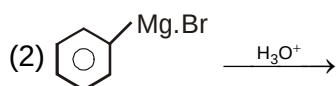
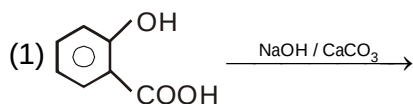
77. $\text{MeCH}_2\text{C}\equiv\text{CH} \xrightarrow{\text{NH}_3 / \text{NaNH}_2} \text{A} \xrightarrow{\text{Et Br}} \text{B}$,
A and B are

- (1) $\text{MeCH}_2\text{C}\equiv\text{C}-\text{Na}$, $\text{MeCH}_2\text{C}\equiv\text{C}-\text{Et}$
- (2) $\text{MeCH}_2\text{CH}=\text{CH}_2$, $\text{MeCH}_2-\text{CHEt}-\text{CH}_3$
- (3) $\text{MeCH}_2\text{CH}=\text{CHNH}_2$, $\text{MeCH}_2\text{CH}=\text{CH}-\text{NHBr}$
- (4) $\text{MeCH}_2\text{C}\equiv\text{C}-\text{NH}_2$, $\text{MeC}\equiv\text{C}-\text{NH}-\text{Br}$

78. Propyne and HCl in presence of peroxide gives :

- (1) 1, 1-dichloro propane
- (2) 1, 2-dichloro propane
- (3) 1, 3-dichloro propane
- (4) 2, 2-dichloro propane

79. In which of the following reactions benzene is obtained



80. The active species in the nitration of benzene is

- (1) NO_2^+
- (2) HNO_3
- (3) NO_3^-
- (4) NO_2^-

81. The function of anhydrous AlCl_3 in the Friedel craft's reaction

- (1) To absorb water
- (2) To absorb HCl
- (3) To produce electrophile
- (4) To produce Nucleophile

82. Which of the following reaction shows that there are three C=C bond in benzene

- (a) $\text{C}_6\text{H}_6 + \text{Br}_2 \xrightarrow{\text{FeBr}_3}$
- (b) $\text{C}_6\text{H}_6 + \text{HNO}_3 \xrightarrow{\text{H}_2\text{SO}_4}$
- (c) $\text{C}_6\text{H}_6 + 3\text{O}_3 \longrightarrow$
- (d) $\text{C}_6\text{H}_6 + 3\text{H}_2 \xrightarrow{\text{Ni}}$

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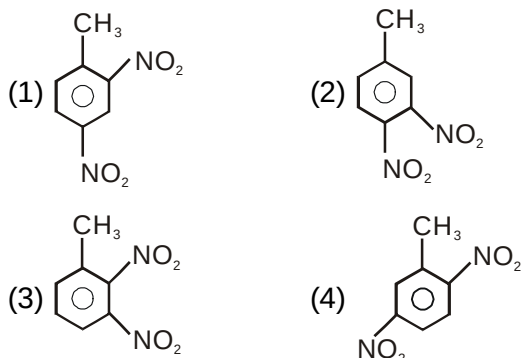
- (1) a, c
- (2) b, d
- (3) b, c, d
- (4) c, d

83. Chlorination of toluene in the presence of light and heat followed by treatment with aqueous NaOH gives

- (1) o-cresol
- (2) p-cresol
- (3) 2, 4-dihydroxy toluene
- (4) Benzyl alcohol

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84. p-nitro toluene on further nitration gives



85. $A \xleftarrow{Br_2} \text{C}_6\text{H}_5\text{CH=CH}_2 \xrightarrow{KMnO_4} B$; A and B respectively are

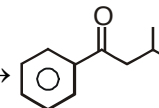
- (1) o-bromo styrene, benzoic acid
(2) p-bromostyrene, benzaldehyde
(3) m-bromostyrene, benzaldehyde
(4) Styrene dibromide, benzoic acid.

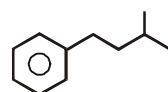
86. Which of the following reaction gives t-butyl benzene:

- (1) $C_6H_6 + \text{t-butyl chloride} \xrightarrow{AlCl_3}$
(2) $C_6H_6 + (CH_3)_2C=CH_2 \xrightarrow{HF}$
(3) $C_6H_6 + \text{t-butyl alcohol} \xrightarrow{H_2SO_4}$
(4) All of them

87. The ozonolysis product of 1, 2-dimethyl benzene is / are

- (1) Only $CH_3-C(=O)-C(=O)-CH_3$
(2) $CH_3-C(=O)-C(=O)-H + CH_3-C(=O)-C(=O)-CH_3$
(3) $H-C(=O)-C(=O)-H + CH_3-C(=O)-C(=O)-CH_3$
(4) $CH_3C(=O)C(=O)CH_3 + HC(=O)CH(=O) + CH_3C(=O)C(=O)H$

88. Benzene $\xrightarrow{A}$  $\xrightarrow{B}$

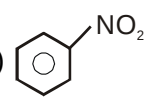
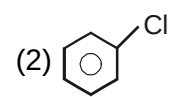
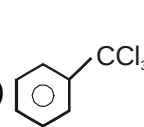
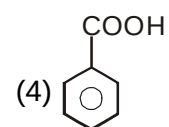
 ; A and B respectively are

- (1) $Zn-Hg + HCl$, $CH_3CH_2COCl + AlCl_3$
(2) $CH_3CH_2COCl + AlCl_3$, $LiAlH_4$
(3) $CH_3CH_2COCl + AlCl_3$, $NaBH_4$
(4) $CH_3CH_2COCl + AlCl_3$, $Zn-Hg + HCl$

89. Which of the following reaction does not give benzoic acid

- (1) $C_6H_5-CH_3 \xrightarrow{Acidic KMnO_4}$
(2) $C_6H_5-CH_3 \xrightarrow{CrO_2Cl_2, H_2O} \xrightarrow{[O]}$
(3) $C_6H_5-C\equiv CH \xrightarrow{Acidic KMnO_4}$
(4) $C_6H_5-C(CH_3)_2-CH_3 \xrightarrow{Acidic KMnO_4}$

90. In which of the following compound the electrophile attack on o- and p- positions :

- (1)  (2) 
(3)  (4) 

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91. Most of the transgenic animals are:
(1) Mice (2) Cows
(3) Pigs (4) Rabbits
92. Genetically modified rice is rich in:
(1) Vitamin A (2) Vitamin B₁₂
(3) Vitamin D (4) Vitamin E
93. Select the incorrect statement :-
(1) Bt cotton is resistant from the infection of *Bacillus thuringiensis*.
(2) During green revolution food production was tripled.
(3) cry genes code for Bt toxins.
(4) Genetically modified rice is rich in Vitamin A.

94. Match the columns :

Column A		Column B	
A.	Gene modification	1	Vitamin A enriched rice
B.	Gene therapy	2	Chemical safety testing
C.	Transgenic animals	3	ADA deficiency treatment
D.	Bt toxin	4	Cry gene

- (1) A-1, B-3, C-2, D-4
(2) A-4, B-3, C-2, D-1
(3) A-1, B-2, C-3, D-4
(4) A-4, B-2, C-3, D-1
95. Which is the source of environmental pollution?
(1) Agrochemicals
(2) Genetically modified crops
(3) Both (1) & (2)
(4) None of these
96. Which gene controls the corn borers?
(1) Cry I Ac (2) Cry II Ab
(3) Cry I Ab (4) Both (2) and (3)
97. Which of the following methods involves silencing of mRNA?
(1) RNA transcription
(2) RNA interferences
(3) DNA replication
(4) DNA fingerprinting
98. For the first time human insulin was prepared in :
(1) 1972 (2) 1983
(3) 1990 (4) 1997

99. First clinical gene therapy was given to a patient in :-
(1) 1983 (2) 1990
(3) 1987 (4) 2003
100. Permanent cure for ADA (Adenosine Deaminase) deficiency is :-
(1) Blood transfusion
(2) Enzyme replacement therapy
(3) Introduction of gene producing ADA in early embryo
(4) Bone marrow transplantation
101. Incorrect about PCR (Polymerase Chain Reaction) is :-
(1) It makes the copies of DNA.
(2) It is based upon antigen antibody reaction
(3) It can detect HIV
(4) It can detect cancer
102. Which part of tobacco plant is infected by *Meloidogyne incognita*?
(1) Stem (2) Root
(3) Leaf (4) Flower
103. Which of the following is first transgenic cow?
(1) Andi (2) Rosie
(3) Polly (4) Lucy
104. Use of biological resources by the multinational companies without proper authorization from the concerned countries and people without compensatory payment is :
(1) Biopatent
(2) Biopiracy
(3) Biowar
(4) Bioterrorism
105. "A probe is allowed to hybridize with its complimentary DNA in a clone of cells and is later on detected by autoradiography." Above step is the part of :
(1) Biopiracy
(2) Polymerase chain reaction
(3) RNA interference
(4) Gene therapy

106. Match the columns :

Column A		Column B	
A.	Biopatent	1	E. coli
B.	Human Insulin	2	Gene Therapy
C.	Pest resistant Tobacco plants	3	Genetic Engineering Approved Committee
D.	ADA deficiency	4	Transgenic Animals
D.	Treatment of emphysema	5	RNA interference

(1) A-3, B-1, C-5, D-2, E-4

(2) A-1, B-3, C-4, D-5, E-2

(3) A-3, B-4, C-5, D-2, E-1

(4) A-5, B-2, C-1, D-3, E-4

107. Leaf mesophyll cell is part of :-

(1) Epidermal tissue system

(2) Ground tissue system

(3) Vascular tissue system

(4) Protoderm

108. Match the columns

Column A		Column B	
A.	Bt toxin	1	Silencing of m-RNA
B.	RNA interference	2	ADA cDNA
C.	RDT	3	cry gene
D.	Gene Therapy	4	DNA or RNA probe

(1) A-1, B-3, C-2, D-4

(2) A-3, B-1, C-2, D-4

(3) A-3, B-1, C-4, D-2

(4) A-2, B-1, C-3; D-4

109. In a water body, type of benthic animals depends on the :

(1) Sediment

(2) Planktons

(3) pH of water

(4) Salinity of water

110. Birds and mammals are respectively :

(1) Thermoconfirmers and osmoconfirmers

(2) Thermoconfirmers and osmoregulators

(3) Thermoregulators and osmoconfirmers

(4) Thermoregulators and osmoregulators

111. Most of invertebrates are :

(1) Osmoregulators

(2) Thermoconfirmers

(3) Osmoregulators and thermoconfirmers

(4) Thermoregulators

112. During unfavourable conditions, zooplankton species in lakes and ponds undergo a stage of suspended development, known as :

(1) Dormancy

(2) Aestivation

(3) Diapause

(4) Hibernation

113. Plants and most of aquatic animals are respectively :

(1) Osmoconfirmers and osmoregulators

(2) Osmoconfirmers and osmoconfirmers

(3) Osmoregulators and osmoconfirmers

(4) Osmoregulators and osmoregulators

114. Incorrect about thermoregulation:

(1) Expensive for small sized animals

(2) More expensive for animals with low body surface area: volume ratio

(3) It is done in summer months by sweating

(4) It is done by muscle contraction during winter

115. Small animals are rarely found in polar regions because :

(1) Their surface area is small

(2) Their surface area and body mass ratio is low

(3) Their surface area and body mass ratio is high

(4) Large sized animals can feed on them

116. Many birds from Siberia move to this national park / sanctuary in India?

(1) Silent valley national park, Kerala

(2) Chilka lake bird sanctuary, Orissa

(3) Sultanpur lake bird sanctuary, Haryana

(4) Keoladeo national park, Bharatpur, Rajasthan

117. Statement not applicable to habitat is:

- (1) It results from a local and regional variation within a biome
- (2) It consists of physico-chemical components
- (3) It is influenced by biotic components
- (4) It is having optimum conditions for survival but not for reproduction

118. Identify the correct statement :

- (1) Most of the organisms can resist wide variation in temperature
- (2) Most of the aquatic organisms can resist wide variation in salinities
- (3) Most of the organisms can't retain their body temperature constant
- (4) None of these

119. In colder climate, mammals possess shorter ears and limbs to minimise heat loss. This is known as :-

- (1) Allen's Rule
- (2) Bergman's Rule
- (3) Gloger's Rule
- (4) Lindeman's Rule

120. What do the desert lizards do when their body temperature drops?

- (1) They hide in the soil
- (2) They move to shades
- (3) They bask in sun
- (4) They go to sleep

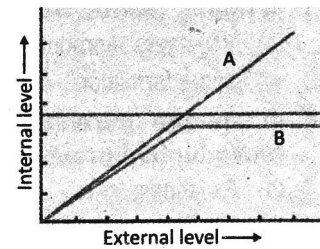
121. Quiescent centre found in :-

- (1) Leaf apex
- (2) Root apex
- (3) Dicot stem
- (4) Stem and leaf

122. Death rate and birth rate are the terms applicable to :

- (1) Population
- (2) Individual
- (3) Biological community
- (4) Ecosystem

123. Diagrammatic representation of organismic response is given. Select the correct option



	A	B
(1)	Confirmers	Regulators
(2)	Regulators	Partial regulators
(3)	Confirmers	Partial, regulators
(4)	Confirmers	Partial confirmers

124. Barnacles grow on the back of whales. Similar type of relationship can be observed in :

- (1) Clown fish and sea anemones
- (2) Cuckoo and koel
- (3) Fig trees and wasp
- (4) Orchids and bees

125. Cuscuta growing on hedge plant is an example of :

- (1) Epiphyte
- (2) Saprophyte
- (3) Parasite
- (4) Competitor

126. In an age pyramid of human population, if pre-reproductive group is less than reproductive group, then it indicates population is :

- (1) Expanding
- (2) Static
- (3) Declining
- (4) Can't interpret

127. In Verhulst-Pearl logistic growth, following equation was derived :

$$\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right)$$

What does K refer to?

- (1) A constant
- (2) Intrinsic rate of natural increase
- (3) Population density
- (4) Carrying capacity

128. Few interactions are given following. How many of them are not examples of mutualism?

- (1) Fungus and roots of higher plants—mycorrhiza
- (2) Clown fish and sea anemones
- (3) Cuckoo and crow
- (4) Orchids and bumblebees
- (5) Cuscuta and Hedge plants
- (6) Sparrow eating seeds

- | | |
|-------|-------|
| (1) 2 | (2) 3 |
| (3) 4 | (4) 5 |

129. Mac Arthur observed that 5 species of Warblers (birds) can coexist on same Fig tree due to behavioural differences in their foraging activities. This is an example of :-

- (1) Resource partitioning
- (2) Competitive release
- (3) Competitive exclusion
- (4) Brood parasitism

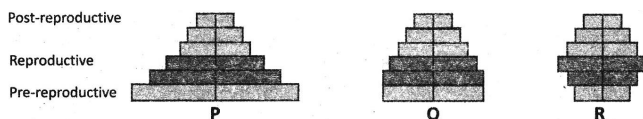
130. Connell's elegant field experiment is illustration for :-

- (1) Competitive inhibition
- (2) competitive release
- (3) coexistence
- (4) biological control

131. What is the magnitude of salinity in inland water?

- (1) < 5ppm (2) 10-15 ppm
(3) 30-35 ppm (4) > 50ppm

132. Various age pyramids (P-R) for human population are shown below. Select the option with correct identification.

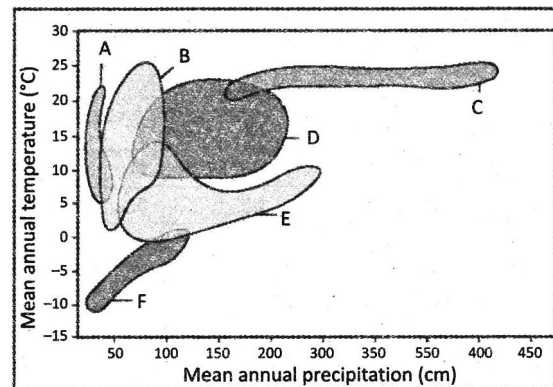


Stable type		Declining type
(1)	P	Q
(2)	P	R
(3)	Q	R
(4)	Q	P

133. In which plant, one of the petals of flower mimics female bee?

- (1) Ophrys (2) Calotropis
(3) Parthenium (4) Lantana

134. Biome distribution is shown with respect to annual temperature and precipitation. Correctly identify the biomes.



Temperate		Grasslands	Arctic and Alpine tundra
(1)	D	B	F
(2)	C	E	A
(3)	E	B	F
(4)	F	A	C

135. Read the statements (A-E) and answer the question following them.

- (A) Permanent tissues in plants do not have the capability of division.
- (B) After grazing by herbivores, grasses regenerate their lost parts by the action of intercalary meristem.
- (C) Interfascicular cambium is an example of apical meristem.
- (D) Metaxylem is a type of primary xylem.
- (E) Vessels are found in almost all seed plants.
- How many statements are false?

- (1) 1 (2) 2 (3) 3 (4) 4

136. Axillary bud develops from :-

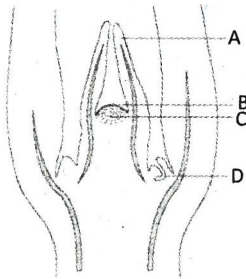
- (1) Shoot apical meristem
- (2) Root apical meristem
- (3) Intercalary meristem
- (4) Mature cells

137. How many of the following are examples of lateral meristem?

[Cork cambium, Intercalary meristem, intrafascicular cambium, interfascicular cambium, Apical meristem]

(1) 4 (2) 3 (3) 2 (4) 1

138. Section of shoot apex is shown in the diagram. Select the option with correct identification of labelling and one of its features.

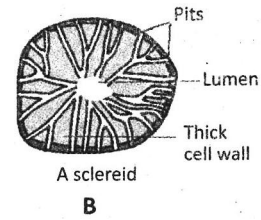
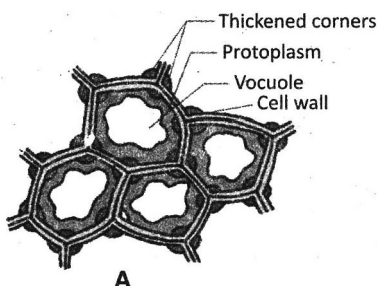


- (1) A–Plumule, gives rise to shoot
(2) B–Shoot apex, possess apical meristem
(3) C–Meristematic zone, cells are actively dividing giving rise to trichomes
(4) D–Leaf primordium, give rise to leaves and lateral branches

139. Which permanent tissue is incorrectly matched with its cell wall component and function?

(1)	Collenchyma	Cellulose, hemicelluloses and pectin	Mechanical support
(2)	Sclerenchyma	Lignin	Mechanical support
(3)	Xylem vessel	Suberin	Mechanical support and conduction of water
(4)	Parenchyma	Cellulose	Photosynthesis and storage

140. Diagram of some simple permanent tissues are given (A and B). Select the option with correct information about A and B :

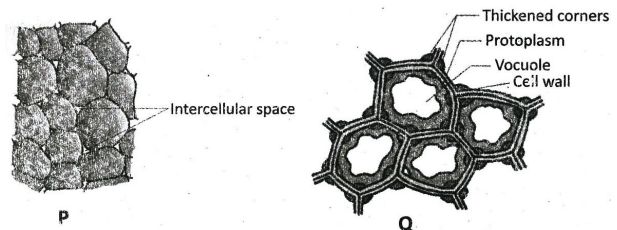


	A	B
(1)	Dead cells	Living cells
(2)	Provide mechanical strength	Mainly photosynthetic in action
(3)	Devoid of protoplasm	Lignified cell wall
(4)	Cell wall thickened at corners	Protoplasm absent

141. Which of the following provides mechanical support to the growing parts of plants like young stem and petiole of a leaf?

- (1) Sclerenchyma (2) Collenchyma
(3) Parenchyma (4) Xylem vessels

142. P and Q are two types of simple permanent tissues. Select the correct option with their occurrence :



	P	Q
(1)	Cortex of dicot stem	Below epidermis in dicot stem
(2)	Pulp of guava	Pith in monocot root
(3)	Medullary rays	Epidermis of stem
(4)	Xylem and phloem	Lignified fibres

143. Sclereids are found in all of the following except :

- (1) Solid endosperm of coconut
- (2) Fruit wall of nuts
- (3) Seed coat of legumes
- (4) Pulp of fruits like pear, guava and sapota

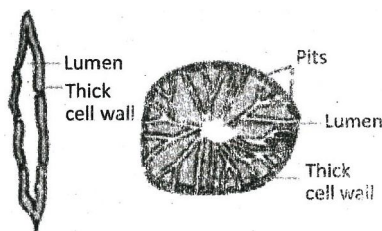
144. Parenchyma and xylem, both are examples of :

- (1) Simple tissue
- (2) Complex tissue
- (3) Permanent tissue
- (4) Secondary meristematic tissue

145. Cell wall of sclerenchymatous fibres and sclereids are respectively thickened by :

- (1) Cellulose, cellulose
- (2) Cellulose, hemicelluloses
- (3) Lignin, cellulose
- (4) Lignin, lignin

146. Observe the diagrams and select the option with their incorrect occurrence :



- (1) Hypodermis of dicot stem
- (2) Leaves of tea
- (3) Pulp of sapota
- (4) Fruit wall of nuts

147. What is not common in tracheids and vessels?

- (1) Both are dead
- (2) Lignified cell walls
- (3) Structure
- (4) Absence of protoplasm

148. Nucleus of cell A controls the activities of cell B in secondary phloem. A and B are respectively :-

- (1) Phloem parenchyma, phloem fibres
- (2) Bast fibres, companion cells
- (3) Companion cell, sieve tube
- (4) Sieve tube, companion cell

149. Identify the structure given and select the option with incorrect information regarding it :

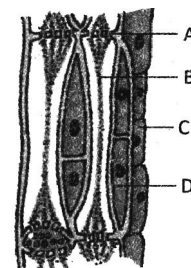


- (1) Given cells are dead in nature
- (2) Walls are heavily suberised
- (3) Component of secondary hadrome
- (4) Perform conduction of sap especially in gymnosperms

150. Root hairs and trichomes are respectively

- (1) Unicellular and unicellular
- (2) Unicellular and multicellular
- (3) Multicellular and unicellular
- (4) Multicellular and muticellular

151. In the given diagram, identify at last two of the labellings (A-D) :



- (1) A-Pits, C-Hypodermis
- (2) A-Sieve tube, B-Tracheid
- (3) B-Sieve pore, C-Companion cell
- (4) B-Sieve tube element, D-Companion cell

152. In grasses, shape of guard cells in stomata is :

- (1) Dumb-bell shaped
- (2) Bean shaped
- (3) Spindle shaped
- (4) Pyriform shaped

153. Which of the following is not the part of ground tissue system?

- (1) Vascular bundles
- (2) Pericycle
- (3) Pith
- (4) Cortex

154. Typical dicot stem possess open vascular bundles. What is incorrect about such vascular bundles?
- Inter-fascicular cambium is present between xylem and phloem
 - It may show secondary growth
 - It may give rise to secondary xylem and secondary phloem both
 - Cambium of such bundles is example of secondary meristem
155. During secondary growth in dicot stem, cambial ring is formed by intrafascicular cambium and:
- Medullary rays
 - Phellogen
 - Interfascicular cambium
 - Phellem
156. In epidermal tissue system, what is incorrect about an epidermal cell?
- Collenchymatous cell with abundant cytoplasm
 - Collenchymatous cell with little cytoplasm
 - Parenchymatous cell with abundant cytoplasm
 - Parenchymatous cell with little cytoplasm
157. Identify A and B. Select the option with their correct occurrence.:



A		B
(1)	Gymnosperms	Gymnosperms and angiosperm
(2)	Gymnosperms only	Angiosperms
(3)	Angiosperms	Gymnosperms only
(4)	Gymnosperms and angiosperms	Angiosperm only

158. Bast fibres are P in primary phloem and Q in secondary phloem.
- P=absent, Q = few
 - P = absent, Q = abundant
 - P = abundant, Q = few
 - P = abundant, Q = abundant
159. Diagram of a conducting tissue is given, in which plants this structure can be observed?



- Angiosperms
 - Gymnosperms
 - Bryophytes
 - Pteridophytes
160. Closed vascular bundles lack:
- Xylem
 - Phloem
 - Cambium
 - None of these
161. Which of the following is not correctly matched with its feature?
- Trichomes—Multicellular
 - Guard cells in dicots—Bean-shaped
 - Hemp fibre—Bast fibre
 - Cork cambium—Secondary growth in stellar region
162. Location of sclerenchyma is :-
- Pulp of guava
 - Pulp of mango
 - Endosperm of coconut
 - Pith of dicot stem
163. In primary xylem of dicot stem, protoxylem is located towards1..... and metaxylem is towards2..... This condition is known3.....
- centre, periphery, exarch
 - centre, periphery, endarch
 - periphery, centre, exarch
 - periphery, centre, endarch

164. Out of the four components of xylem—tracheids, vessels, xylem fibres and xylem parenchyma, how many are dead?

- (1) None (2) One
(3) Two (4) Three

165. On the basis of following features, identify the xylem element.

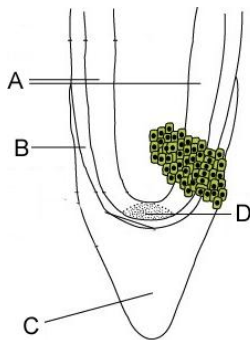
1. Tubular structure
2. No tapering ends
3. Dead cells
4. Absent in gymnosperms

- (1) Tracheids
(2) Vessels
(3) Xylem fibres
(4) Xylem parenchyma

166. Which of the following elements of xylem lack lignified cell wall?

- (1) Xylem parenchyma
(2) Xylem fibres
(3) Vessels
(4) Both (1) and (2)

167. Section of root apex is given. Identify the incorrect information about labelling given in diagram :-

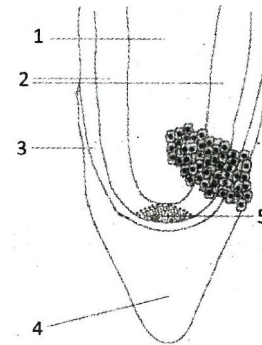


- (1) B gives rise to epidermis of root and also to root hairs
(2) C is protective for root apex
(3) A possess vascular bundles
(4) D is group of actively dividing cells

168. Which of the following feature is unrelated to collenchyma in plants?

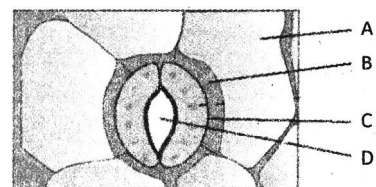
- (1) Cell wall thickening—cellulose, hemicellulose and pectin
(2) intercellular space—developed
(3) Main function—mechanical support
(4) Chloroplast—may or may not be present

169. Section of root is given. Select the option with correct identification:



- (1) 1 -pith, 3-Intercalary meristem
(2) 2-cortex, 5-root apical meristem
(3) 1-central cylinder, 5-lateral meristem
(4) 3-protoderm, 4-root apical meristem

170. A typical dicot stomata is shown in the diagram with labelling A-D. Which labelling is not correctly matched with its feature / function?

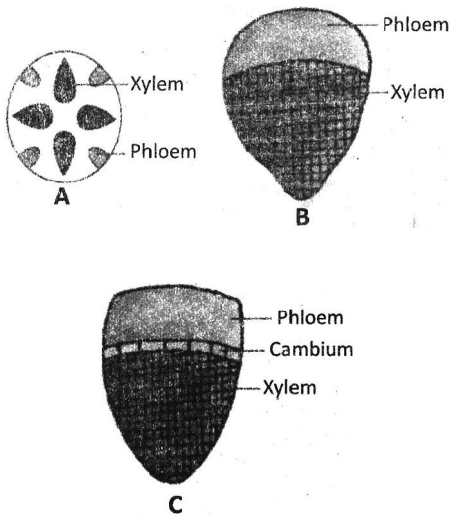


- (1) B—These are specialized epidermal cells
(2) A — Parenchymatous cell with abundant cytoplasm
(3) D — Opening and closing is related with K^+ and ABA
(4) C — Inner wall is thicker than outer wall

171. Cork is also known as:

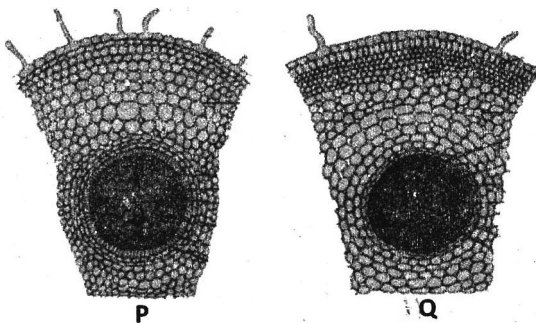
- (1) Phellem
(2) Phellogen
(3) Phelloderm
(4) Bark

172. In the given diagrams, 3 types of vascular bundles are shown. Which of them are found in dicots?



- (1) A and B (2) Only C
(3) A and C (4) A, B and C

173. Observe the sections given (P and Q) and correctly identify their occurrence in the option given:



	P	Q
(1)	Monocot stem	Dicot Stem
(2)	Monocot root	Dicot Stem
(3)	Dicot root	Monocot root
(4)	Monocot root	Dicot root

174. During secondary growth in dicot stem, cork cambium has cut off few cells which are impermeable to water. These cell have been cut off towards:

- (1) Outer side
(2) Inner side
(3) Lateral side
(4) In all directions

175. Differential activity of cambium is observed in :-

- (1) Stem in tropical region
(2) Stem in temperate region
(3) Root in tropical region
(4) Root in temperate region

176. Which type of wood is incorrectly matched with its given feature?

- (1) Heartwood—Provides mechanical support
(2) Spring wood—Possess vessels with wider lumen
(3) Autumn wood—Darker in appearance
(4) Sapwood—Functionally inactive

177. Select the option with incorrect information regarding dicot stem:

- (1) Hypodermis—Possess collenchymatous cells
(2) Endodermis—Also known as starch sheath
(3) Medullary rays—Made up of parenchymatous cells
(4) Pith—Poorly developed or absent

178. Vascular bundles in dicot stem are :-

- (1) Conjoint, open and exarch
(2) Conjoint, open and endarch
(3) Conjoint, closed and exarch
(4) Conjoint, closed and endarch

179. Radial vascular bundles with polyarch condition are found in :-

- (1) Dicotstem
(2) Dicotroot
(3) Monocot stem
(4) Monocotroot

180. Which weed produces highly toxic cardiac glycosides?

- (1) Parthenium
(2) Calotropis
(3) Orchids
(4) Eichhornia